



**Technical School of
Telecommunications Engineering
and Systems**



ANTENNAS AND EMC

SOLUTION

**LABORATORY PRACTICE 2
WIRE ANTENNA ANALYSIS**

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1. Introduction

This laboratory exercise covers the study, design, and analysis of antenna arrays and Yagi antennas. Its main purpose is to apply the theoretical concepts of radiation, mutual impedance, directivity, and radiation patterns to wire-antenna design using full-wave electromagnetic simulation based on the method of moments.

The first part addresses the design of an eight-dipole broadside array with optimum Dolph-Chebyshev excitation to control the sidelobe level. The mutual-impedance matrix, the radiation pattern in the transverse plane, sensitivity to $\pm 5\%$ operating-frequency variations, and beam steering toward non-broadside directions ($\phi = 0^\circ$ and 45°) are analyzed.

The second part develops a four-element Yagi antenna consisting of one reflector, a folded-dipole driven element, and two directors. The target specifications are directivity above 9 dB, a front-to-back ratio above 15 dB, and a purely real input impedance. The antenna bandwidth is then determined from the frequency interval over which both the radiation and impedance specifications are met.

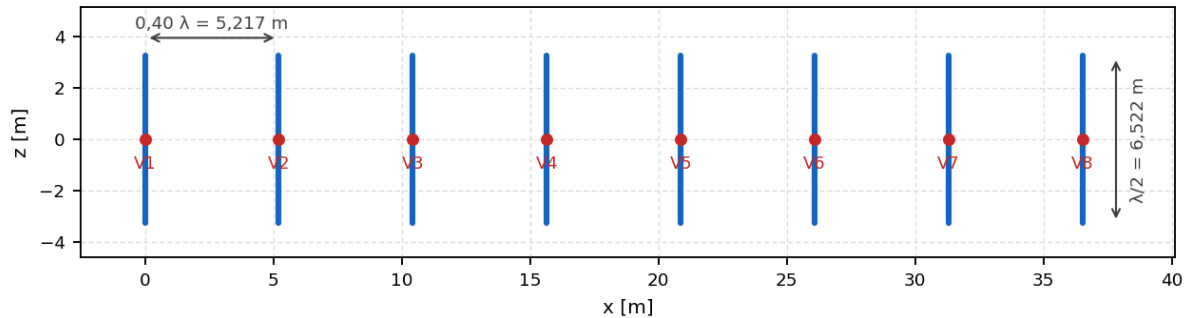
The report presents the design procedures, simulation results (models Prob4A, Prob4B, and Prob5A-Prob5E), and comments for each section.

Note: all analyses use the personalized operating frequency obtained from ID number 02757023J: 23 MHz, with $\lambda = 13.0435$ m, wire diameter $d = \lambda/207 = 63.01$ mm, and dipole spacing $0.40\lambda = 5.2174$ m.

2. DOLPH-CHEBYSHEV

2.1. MUTUAL-IMPEDANCE MATRIX AND S-PARAMETER CALCULATION

The array consists of $N = 8$ dipoles, each $\lambda/2 = 6.5217$ m long and with diameter $d = \lambda/207$. They are placed along the x-axis, parallel to the z-axis, and separated by $0.40\lambda = 5.2174$ m. Each dipole is center-fed by a voltage port (V1 ... V8).



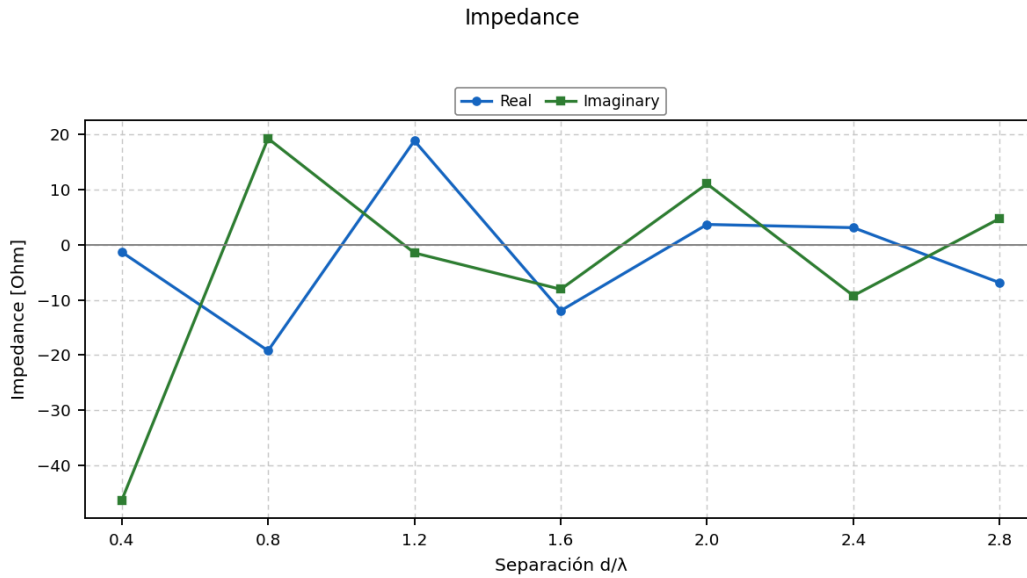
Array de 8 dipolos $\lambda/2$ sobre el eje x ($d = \lambda/207$, puertos en el centro) - Prob4B

The excitation data require the array mutual-impedance matrix. Following the laboratory procedure, the self terms Z_{ii} are obtained by feeding one isolated dipole (Prob4A_ParZii), while Z_{ij} is found by feeding a two-dipole network with the spacing corresponding to pair (i, j) and solving the resulting two-port model (Prob4A_ParZij). At 23 MHz:

$$Z_{ii} = 89,760 + j50,159 \, \Omega$$

	Pair (i, j) dij/λ dij [m] Zij [Ω]		
$ i - j = 1$	0,4	5,217	-1,322 - j46,382
$ i - j = 2$	0,8	10,435	-19,229 + j19,285
$ i - j = 3$	1,2	15,652	18,874 - j1,478
$ i - j = 4$	1,6	20,870	-11,968 - j8,078
$ i - j = 5$	2,0	26,087	3,685 + j11,049
$ i - j = 6$	2,4	31,304	3,105 - j9,241
$ i - j = 7$	2,8	36,522	-6,886 + j4,773

The real and imaginary parts of Z_{ij} alternate in sign and decrease with distance, as expected for radiative coupling between parallel dipoles ($|Z_{12}| = 46.4 \Omega$ at 0.4λ , compared with $|Z_{18}| = 8.4 \Omega$ at 2.8λ).



Mutual impedance Z_{ij} (Frequency = 23 MHz) - Prob4A_ParZij

These values could be used to construct an 8×8 Toeplitz matrix. To include the effect of all eight conductors, including parasitic scattering from unexcited dipoles, the complete eight-port model Prob4A_ParS is also simulated. POSTFEKO is used to export the S-parameter matrix with $Z_0 = 50 \Omega$ in Touchstone format (parametrosS.s8p).

parametrosS.s8p

```
! S-parameter export from POSTFEKO
! Filename: Prob4A_ParS
! Creation date: 2025-11-19 10:42:08
!
# Hz S RI R 50.000000E+00
2.300000E+07 4.0847835E-01 1.8376041E-01 -1.4242752E-01 -1.4242639E-01 2.8576755E-02 6.4760138E-02 2.3190515E-03 -3.9333797E-02
-1.4544389E-02 2.1732870E-02 1.7012488E-02 -7.8454604E-03 -1.3414621E-02 -2.1267135E-03 5.5773953E-03 8.1888323E-03
-1.4242752E-01 -1.4242639E-01 4.3633016E-01 1.2833533E-01 -1.5867514E-01 -1.2811592E-01 4.0444319E-02 6.1947752E-02
-5.7589687E-03 -4.1698505E-02 -1.0049178E-02 2.5919296E-02 1.5445202E-02 -1.1418521E-02 -1.3414621E-02 -2.1267135E-03
2.8576755E-02 6.4760138E-02 -1.5867514E-01 -1.2811592E-01 4.3633016E-01 1.2589410E-01 -1.6326843E-01 -1.2870419E-01
4.3187270E-02 6.3471314E-02 -7.0070104E-03 -4.2952607E-02 -1.0049178E-02 2.5919296E-02 1.7012488E-02 -7.8454604E-03
2.3190515E-03 -3.9333797E-02 4.0444319E-02 6.1947752E-02 -1.6326843E-01 -1.2870419E-01 4.4613540E-01 1.2701118E-01
-1.6437717E-01 -1.2951775E-01 4.3187270E-02 6.3471314E-02 -5.7589687E-03 -4.1698505E-02 -1.4544389E-02 2.1732870E-02
-1.4544389E-02 2.1732870E-02 -5.7589687E-03 -4.1698505E-02 4.3187270E-02 6.3471314E-02 -1.6437717E-01 -1.2951775E-01
4.4613540E-01 1.2701118E-01 -1.6326843E-01 -1.2870419E-01 4.0444319E-02 6.1947752E-02 2.3190515E-03 -3.9333797E-02
1.7012488E-02 -7.8454604E-03 -1.0049178E-02 2.5919296E-02 -7.0070104E-03 -4.2952607E-02 4.3187270E-02 6.3471314E-02
-1.6326843E-01 -1.2870419E-01 4.3633016E-01 1.2833533E-01 -1.5867514E-01 -1.2811592E-01 2.8576755E-02 6.4760138E-02
-1.3414621E-02 -2.1267135E-03 1.5445202E-02 -1.1418521E-02 -1.0049178E-02 2.5919296E-02 -5.7589687E-03 -4.1698505E-02
4.0444319E-02 6.1947752E-02 -1.5867514E-01 -1.2811592E-01 4.3633016E-01 1.2833533E-01 -1.4242752E-01 -1.4242639E-01
5.5773953E-03 8.1888323E-03 -1.3414621E-02 -2.1267135E-03 1.7012488E-02 -7.8454604E-03 -1.4544389E-02 2.1732870E-02
2.3190515E-03 -3.9333797E-02 2.8576755E-02 6.4760138E-02 -1.4242752E-01 -1.4242639E-01 4.0847835E-01 1.8376041E-01
```

The S matrix is converted to an impedance matrix in MATLAB (Section 2.2.2). Comparison of its first row with the pairwise method confirms consistency: $Z_{11} = 88.92 + j52.08 \Omega$ versus $89.76 + j50.16 \Omega$ for the isolated dipole, and $Z_{12} = -0.06 - j46.91 \Omega$ versus $-1.32 - j46.38 \Omega$ for the two-dipole model. The small differences are due to higher-order coupling included only in the complete model. The complete matrix is used below because it represents the impedance actually seen by each source.

2.2. CURRENT-AMPLITUDE CALCULATION FOR EACH DIPOLE

2.2.1. Analytical Method

The number of elements is $N = 8$.

The main-lobe-to-sidelobe ratio is $R_{dB} = 30 + C_4 = 37 \text{ dB}$:

$$R = 10^{(R_{dB}/20)} = 70,7946$$

We calculate $x = x_0$ such that $T_m(x_0) = R$, with $m = N - 1 = 7$:

$$x_0 = \cosh((1/(N-1)) \cdot \operatorname{acosh}(R)) = 1,2609$$

$$\begin{aligned} A_3 &= x_0^7 = 5,0682 \\ A_2 &= (-112 x_0^5 + 112 A_3) / 16 = 13,1639 \\ A_1 &= (56 x_0^3 - 56 A_3 + 20 A_2) / 4 = 22,9328 \\ A_0 &= -7 x_0 + 7 A_3 - 5 A_2 + 3 A_1 = 29,6297 \end{aligned}$$

$$A_3 = x_0^7 = 5,0682$$

$$A_2 = (-112 x_0^5 + 112 A_3) / 16 = 13,1639$$

$$A_1 = (56 x_0^3 - 56 A_3 + 20 A_2) / 4 = 22,9328$$

$$A_0 = -7 x_0 + 7 A_3 - 5 A_2 + 3 A_1 = 29,6297$$

The following amplitude values are obtained (normalized to the maximum):

$$[I] = [0,1710 \ 0,4443 \ 0,7740 \ 1,0000 \ 1,0000 \ 0,7740 \ 0,4443 \ 0,1710]$$

MATLAB is then used to calculate the voltage required at each port:

$$[V] = \{Z\} \times [I]$$

2.2.2. MATLAB

CURRENTS

```
% Parametros DNI (02757023)
C5 = 5; C4 = 7;
N = 8;
R_dB = 30 + C4; % 37 dB
sep = 0.30 + 0.02*C5; % 0.40 (en lambdas)

R = 10^(R_dB / 20);

x0 = cosh( (1/(N-1)) * acosh(R) );

A3 = x0^7;
A2 = (-112 * x0^5 + 112 * A3) / 16;
A1 = (56 * x0^3 - 56 * A3 + 20 * A2) / 4;
A0 = -7 * x0 + 7 * A3 - 5 * A2 + 3 * A1;

I_amplitudes = [A3, A2, A1, A0, A0, A1, A2, A3];
I_normalizadas = I_amplitudes / max(I_amplitudes);

%%RESULTADO
fprintf('Amplitudes de corriente normalizada:\n' );
disp(I_normalizadas.);
```

```
>> corrientes
Amplitudes de corriente normalizada:
0.1710
0.4443
0.7740
1.0000
1.0000
0.7740
0.4443
0.1710
```

IMPEDANCES AND VOLTAGES

```
filename = 'parametrosS.s8p';
sobj = sparameters(filename);

Z = s2z(sobj.Parameters, sobj.Impedance);
freqs = sobj.Frequencies;

Zf = Z(:,:,1); % matriz a f0 = 23 MHz

disp('Matriz de impedancias mutuas (Ohm) ');
disp(Zf);

Zmag = abs(Zf);
Zphase = angle(Zf)*180/pi;

I = [0.1710; 0.4443; 0.7740; 1; 1; 0.7740; 0.4443; 0.1710];
V = Zf * I;
disp('Tensiones resultantes (V) ');
disp(V.);
```

Running the MATLAB code yields the mutual-impedance matrix and the port voltages for the broadside case. The numerical matrix and resulting complex voltages are shown in the original-page panel.

```
Matriz de impedancias mutuas (Ohm)
Columns 1 through 7

88.9156 +52.0765i -0.0629 -46.9071i -20.1987 +21.3265i 21.2176 - 2.7397i -14.5987 - 8.7285i 5.3734 +13.3747i 2.9434 -12.2227i
-0.0629 -46.9071i 88.8698 +54.3813i 0.8756 -47.8352i -21.1767 +21.2407i 21.9343 - 2.0501i -14.9915 - 9.5959i 5.5194 +14.0097i
-20.1987 +21.3265i 0.8756 -47.8352i 88.0307 +54.2245i 1.4094 -47.1983i -21.4402 +20.5314i 22.0240 - 1.5809i -14.9915 - 9.5959i
21.2176 - 2.7397i -21.1767 +21.2407i 1.4094 -47.1983i 87.7950 +53.5558i 1.4863 -46.7832i -21.4402 +20.5314i 21.9343 - 2.0501i
-14.5987 - 8.7285i 21.9343 - 2.0501i -21.4402 +20.5314i 1.4863 -46.7832i 87.7950 +53.5558i 1.4094 -47.1983i -21.1767 +21.2407i
5.3734 +13.3747i -14.9915 - 9.5959i 22.0240 - 1.5809i -21.4402 +20.5314i 1.4094 -47.1983i 88.0307 +54.2245i 0.8756 -47.8352i
2.9434 -12.2227i 5.5194 +14.0097i -14.9915 - 9.5959i 21.9343 - 2.0501i -21.1767 +21.2407i 0.8756 -47.8352i 88.8698 +54.3813i
-8.0048 + 7.1248i 2.9434 -12.2227i 5.3734 +13.3747i -14.5987 - 8.7285i 21.2176 - 2.7397i -20.1987 +21.3265i -0.0629 -46.9071i

Column 8

-8.0048 + 7.1248i
2.9434 -12.2227i
5.3734 +13.3747i
-14.5987 - 8.7285i
21.2176 - 2.7397i
-20.1987 +21.3265i
-0.0629 -46.9071i
88.9156 +52.0765i
```

Tensiones resultantes (V)

```
10.2639 - 0.7540i
32.2600 - 4.9895i
56.3421 - 5.5015i
75.2466 - 7.3027i
75.2466 - 7.3027i
56.3421 - 5.5015i
32.2600 - 4.9895i
10.2639 - 0.7540i
```

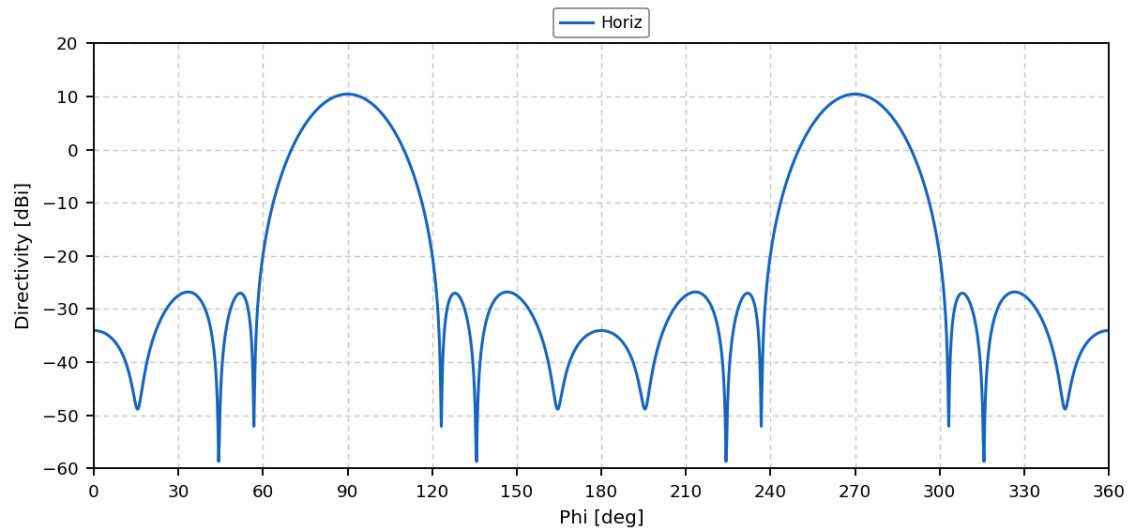
The voltages are symmetric about the center of the array, as are the currents that produce them. The following magnitudes and phases (Amp_n, Afa_n) are entered into the voltage sources of the CADFEKO Prob4B_90 model:

n	Amp _n = V _n [V]	Afa _n = ∠V _n [°]
1	10,2916	-4,2012
2	32,6436	-8,7919
3	56,6101	-5,5770
4	75,6001	-5,5432
5	75,6001	-5,5432
6	56,6101	-5,5770
7	32,6436	-8,7919
8	10,2916	-4,2012

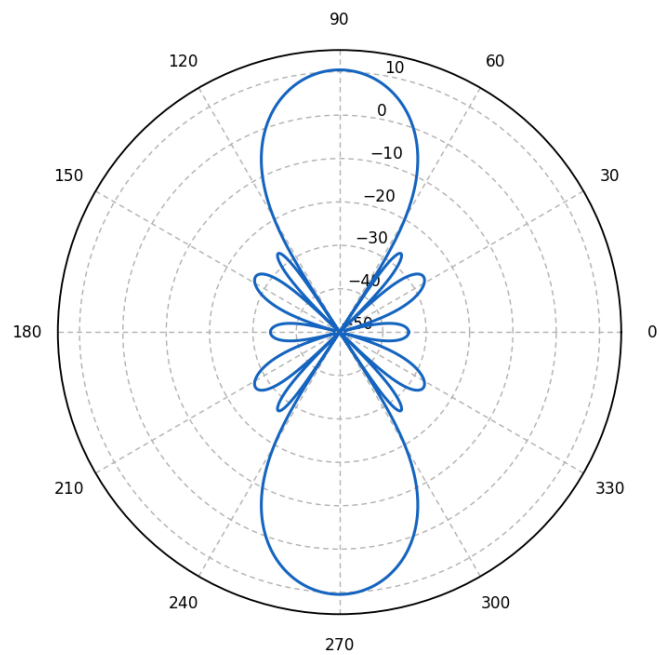
2.3. RADIATION PATTERN IN THE z = 0 PLANE

The complete array is simulated with these voltages (Prob4B_90), and the radiation pattern is generated in the z = 0 plane, corresponding to the $\theta = 90^\circ$ cut. As a verification step, the currents obtained at the ports reproduce the designed Chebyshev distribution with a maximum relative error below 10^{-5} , confirming that the impedance matrix is correct.

Far field



Total Directivity [dBi] (Frequency = 23 MHz; Theta = 90 deg) - Prob4B_90

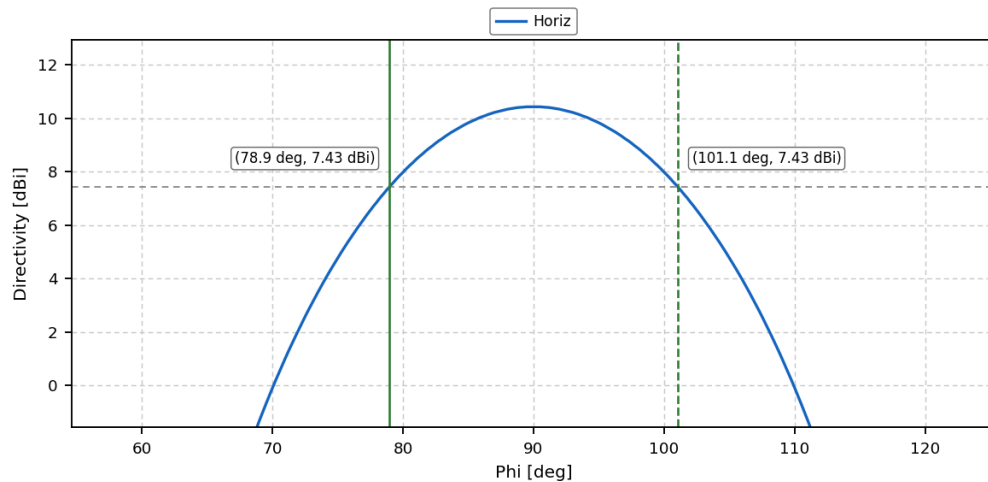


Total Directivity (Frequency = 23 MHz; Theta = 90 deg) - Prob4B_90

Directividad = 10,43 dBi

Zooming into the main lobe and measuring the -3 dB points with cursors gives:

Far field



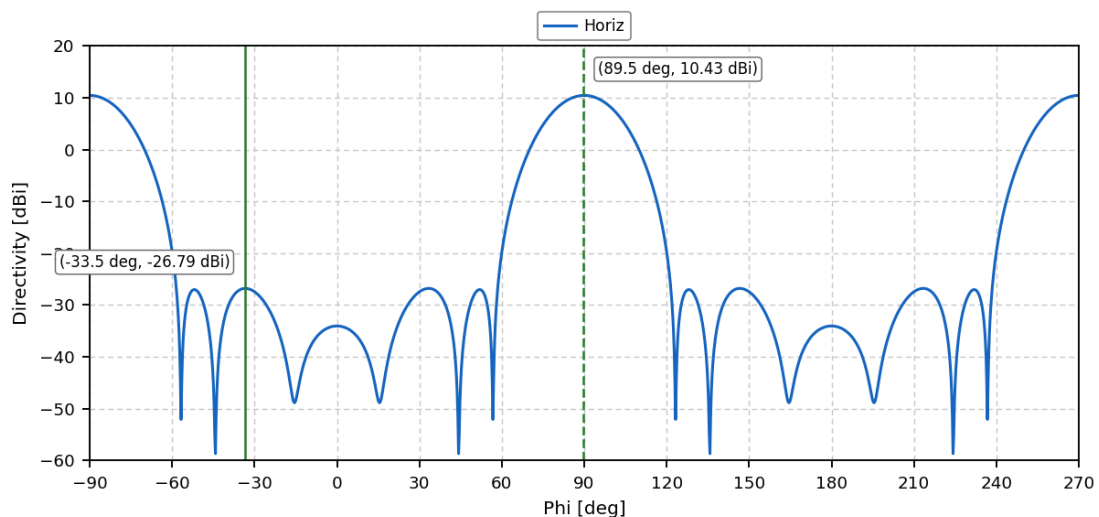
Total Directivity [dBi] (Frequency = 23 MHz; Theta = 90 deg) - Prob4B_90

Half-power beamwidth: $101.05^\circ - 78.95^\circ = 22.1^\circ$

The theoretical array-factor value for $N = 8$, $d = 0.4\lambda$, and $R = 37$ dB is 22.2° , in excellent agreement with the simulation.

The required main-lobe-to-sidelobe ratio is at least 37 dB. The largest sidelobe gives 37.22 dB, so the requirement is met. The constant sidelobe ripple, with all sidelobes at the same level, is also clearly visible and is characteristic of Dolph-Chebyshev synthesis.

Far field



Total Directivity [dBi] (Frequency = 23 MHz; Theta = 90 deg) - Prob4B_90

In this case, the ratio is $-26.79 + 37.22 = 37.22$ dB > 37 dB, so the requirement is met. The constant sidelobe ripple (all sidelobes at the same level), characteristic of Dolph-Chebyshev synthesis, is also observed.

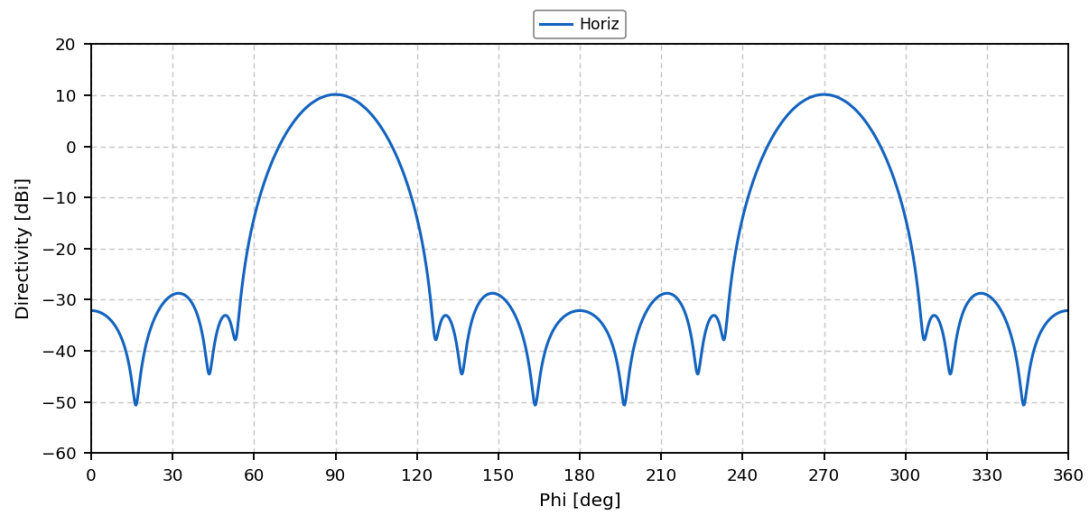
2.4. PARAMETER DEGRADATION FOR $\pm 5\%$ FREQUENCY VARIATION

A three-frequency sweep is performed while keeping the excitation voltages calculated at 23 MHz fixed (Prob4B_90Frec). The array figures of merit degrade away from the design frequency.

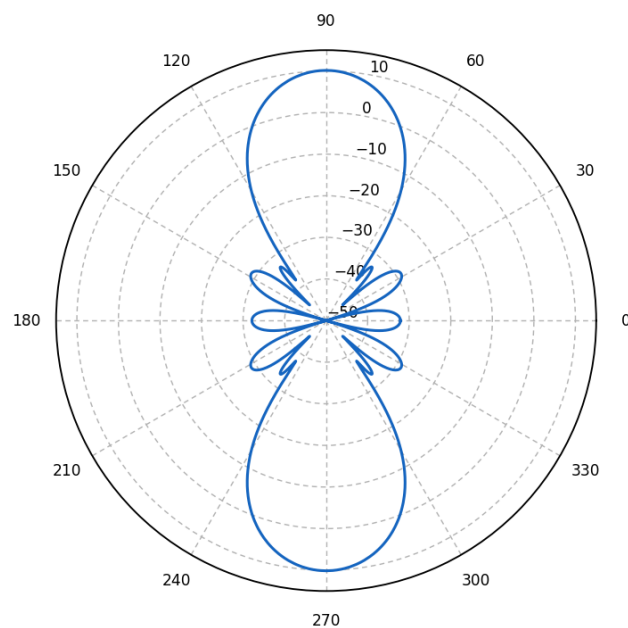
Lower frequency = 5% below the operating frequency:

$$f_{\text{inf}} = 23 \text{ MHz} \times 0,95 = 21,85 \text{ MHz}$$

Far field



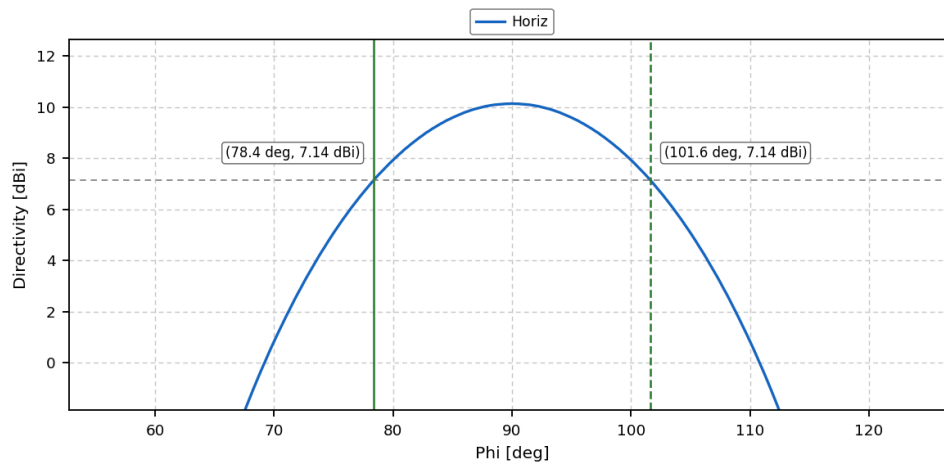
Total Directivity [dBi] (Frequency = 21.85 MHz; Theta = 90 deg) - Prob4B_90Frec



Total Directivity (Frequency = 21.85 MHz; Theta = 90 deg) - Prob4B_90Frec

Directividad = 10,14 dBi

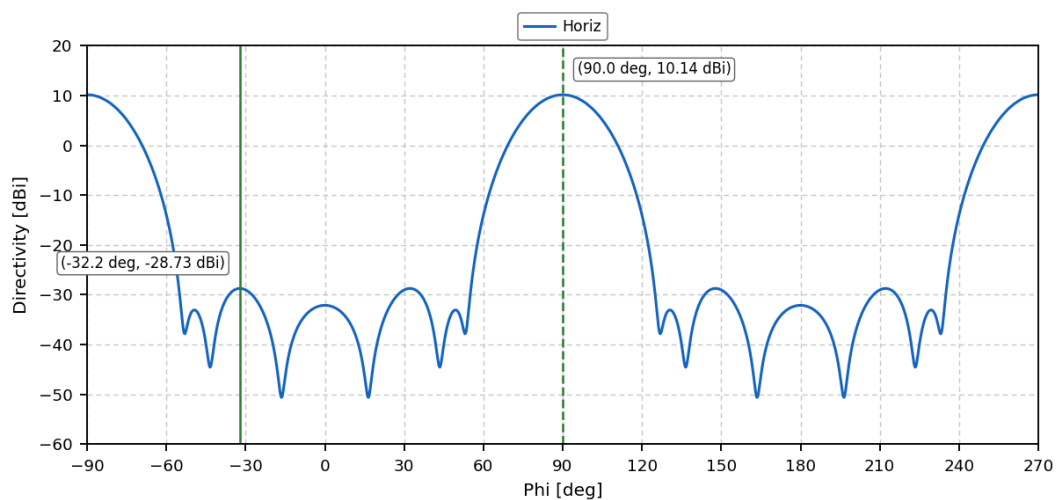
Far field



Total Directivity [dBi] (Frequency = 21.85 MHz; Theta = 90 deg) - Prob4B_90Frec

At 21.85 MHz:

Far field



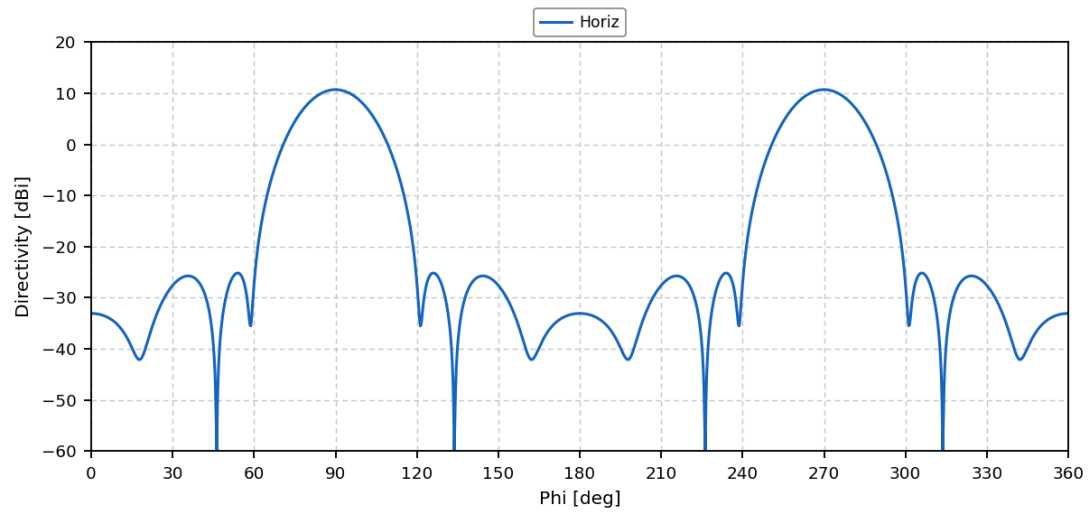
Total Directivity [dBi] (Frequency = 21.85 MHz; Theta = 90 deg) - Prob4B_90Frec

The sidelobe ratio improves slightly at the lower frequency, although directivity decreases and the beam becomes wider.

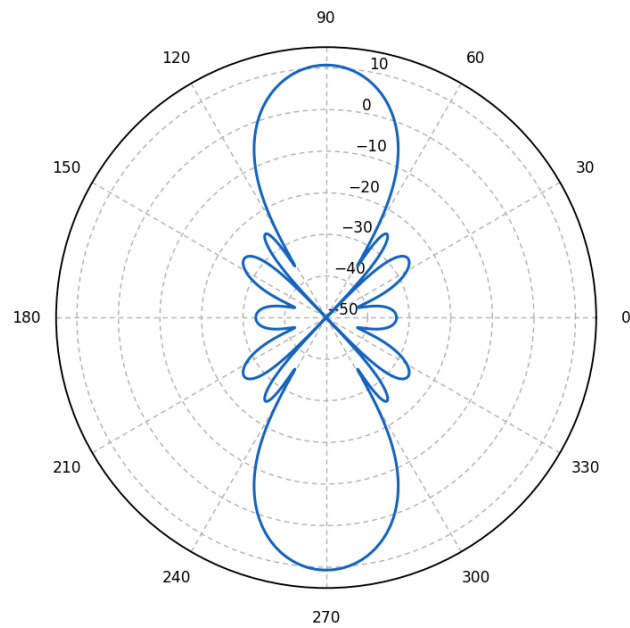
Upper frequency = 5% above the operating frequency:

$$f_{\text{sup}} = 23 \text{ MHz} \times 1,05 = 24,15 \text{ MHz}$$

Far field



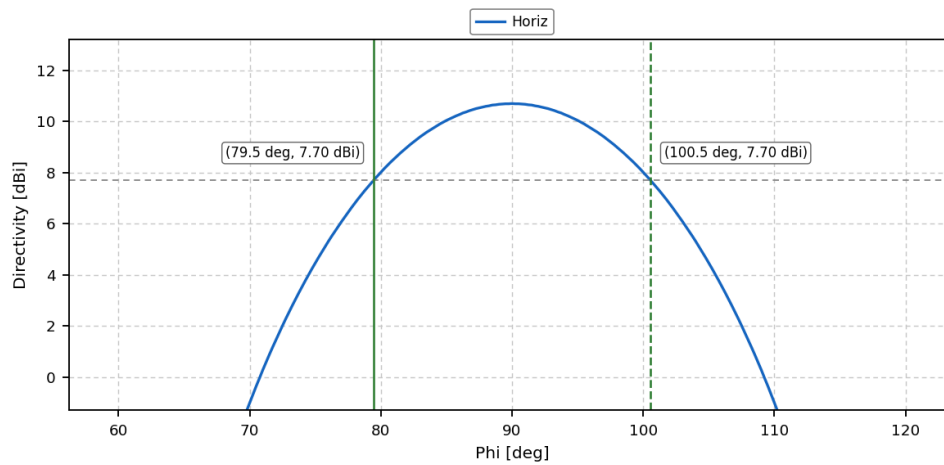
Total Directivity [dBi] (Frequency = 24.15 MHz; Theta = 90 deg) - Prob4B_90Frec



Total Directivity (Frequency = 24.15 MHz; Theta = 90 deg) - Prob4B_90Frec

Directividad = 10,70 dBi

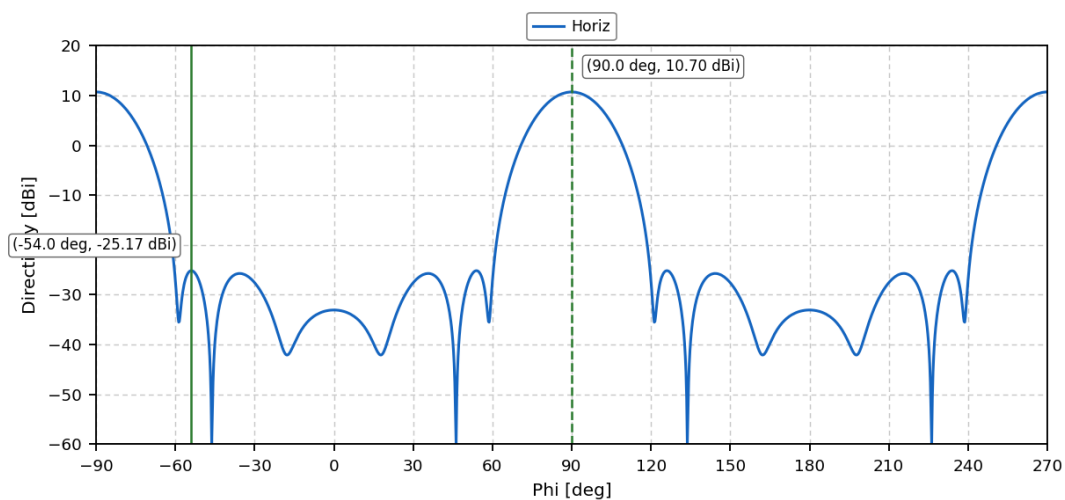
Far field



Total Directivity [dBi] (Frequency = 24.15 MHz; Theta = 90 deg) - Prob4B_90Frec

At 24.15 MHz:

Far field



Total Directivity [dBi] (Frequency = 24.15 MHz; Theta = 90 deg) - Prob4B_90Frec

Therefore, the design specification is not met at the upper test frequency.

2.4.1. STUDY OF THE THREE FREQUENCIES

	Frequency Directivity Beamwidth -3 dB NLPS		
21,85 MHz (-5 %)	10,14 dBi	23,3°	38,87 dB
23 MHz (f_0)	10,43 dBi	22,1°	37,22 dB
24,15 MHz (+5 %)	10,70 dBi	21,1°	35,87 dB

The observed directivity variation is expected: as frequency increases, the electrical length of the array (L/λ) and normalized spacing d/λ increase. The effective aperture is therefore larger, directivity rises, and the beam narrows. The opposite occurs at the lower frequency.

The sidelobe-level behavior is explained by the fact that the generator voltages remain fixed while the impedance matrix changes with frequency. The actual dipole currents are no longer exactly the Chebyshev currents, so the equal-ripple condition is lost. At 24.15 MHz, the taper distortion, together with the reduced visible-region margin caused by the larger d/λ , raises the largest sidelobe above the design level. At 21.85 MHz the opposite occurs and additional margin is obtained. Dolph-Chebyshev synthesis is therefore optimum only at the frequency for which the excitations were calculated.

2.5. Excitations

2.5.1. RADIATION MAXIMUM AT $\phi = 0^\circ$

Beam steering is produced by applying a progressive phase to the currents:
 $\alpha_0 = -k d \cos(\phi_0) = -2\pi(0.40)\cos(\phi_0)$ rad.

```
I = [0.1710; 0.4443; 0.7740; 1; 1; 0.7740; 0.4443; 0.1710];
sep = 0.3 + 5*0.02;

% Para phi0 = 0 grados
phi0 = 0;
alpha0 = -2 * pi * sep * cosd(phi0);
I_scan = I .* exp(1j * (0:7).' * alpha0);
V00 = Zf * I_scan;
disp('Tensiones para phi=0 grados:');
disp(V00.');
```

```
Tensiones para phi=0 grados:
1.0e+02 *

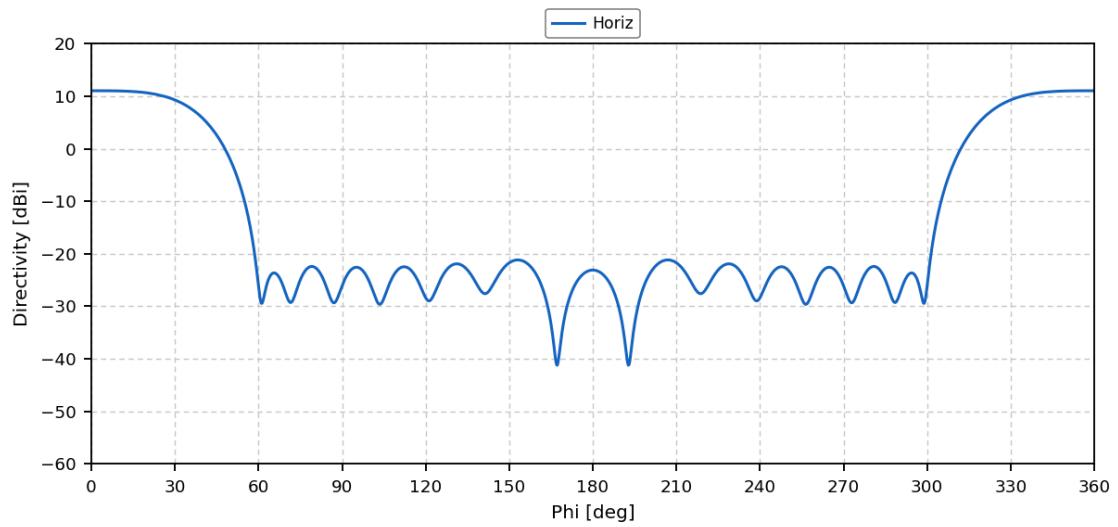
    0.0167 + 0.0652i
    0.0700 - 0.3439i
   -0.5626 + 0.6096i
    1.3186 - 0.3299i
   -1.6188 - 0.6028i
    1.0053 + 1.4801i
    0.0570 - 1.5529i
   -0.7418 + 0.8847i
```

For $\phi_0 = 0^\circ$ (endfire), $\alpha_0 = -144.0^\circ$ between adjacent elements. The MATLAB code is modified accordingly. The following source magnitudes and phases are entered in Prob4B_00:

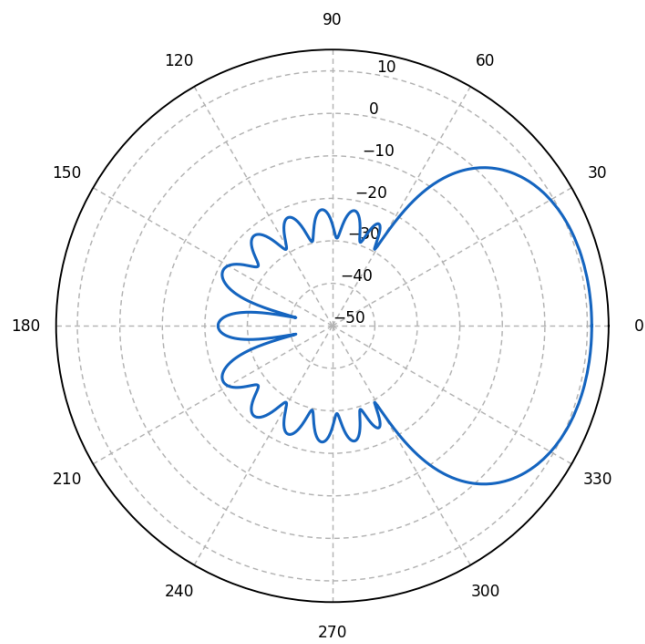
n	$\text{Amp}_n = V_n $ [V]	$\text{Afas}_n = \angle V_n$ [°]
1	6,7328	75,6222
2	35,0949	-78,4926
3	82,9567	132,7018
4	135,9263	-14,0449
5	172,7402	-159,5771
6	178,9193	55,8158
7	155,3948	-87,8995
8	115,4547	129,9778

The radiation pattern obtained in the $z = 0$ plane is:

Far field



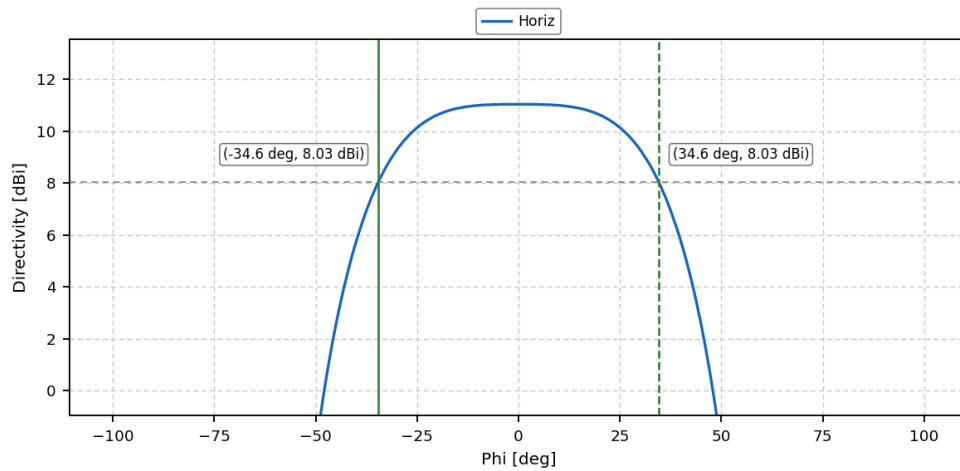
Total Directivity [dBi] (Frequency = 23 MHz; Theta = 90 deg) - Prob4B_00



Total Directivity (Frequency = 23 MHz; Theta = 90 deg) - Prob4B_00

Directividad = 11,03 dBi

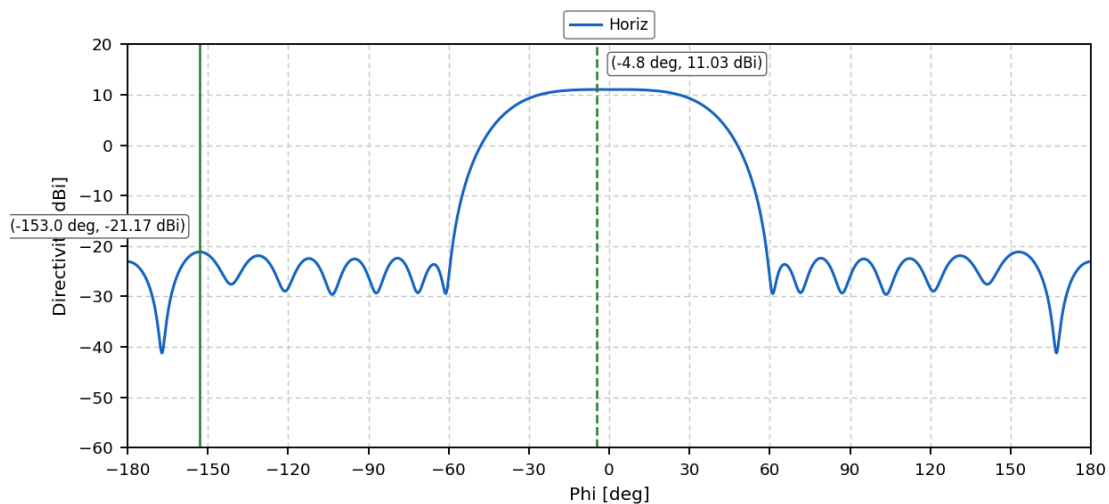
Far field



Total Directivity [dBi] (Frequency = 23 MHz; Theta = 90 deg) - Prob4B_00

Directivity = 11.03 dBi

Far field



Total Directivity [dBi] (Frequency = 23 MHz; Theta = 90 deg) - Prob4B_00

The maximum is correctly located at $\phi = 0^\circ$. The lobe is very flat within approximately $\pm 5^\circ$ of that direction. The largest sidelobe is now the back lobe near $\phi = 180^\circ$, and the main-lobe-to-sidelobe ratio falls to 32.20 dB, below the 37 dB requirement. Thus, the broadside sidelobe specification is no longer met when the array is scanned to endfire.

2.5.2. RADIATION MAXIMUM AT $\phi = 45^\circ$

For $\phi_0 = 45^\circ$, the progressive phase is:
 $\alpha_0 = -2\pi(0.40)\cos(45^\circ) = -101.82^\circ$.

```
I = [0.1710; 0.4443; 0.7740; 1; 1; 0.7740; 0.4443; 0.1710];
sep = 0.3 + 5*0.02;

% Para phi0 = 45 grados
phi0 = 45;
alpha0 = -2 * pi * sep * cosd(phi0);
I_scan = I .* exp(1j * (0:7).' * alpha0);
V45 = Zf * I_scan;
disp('Tensiones para phi=45 grados:');
disp(V45.);
```

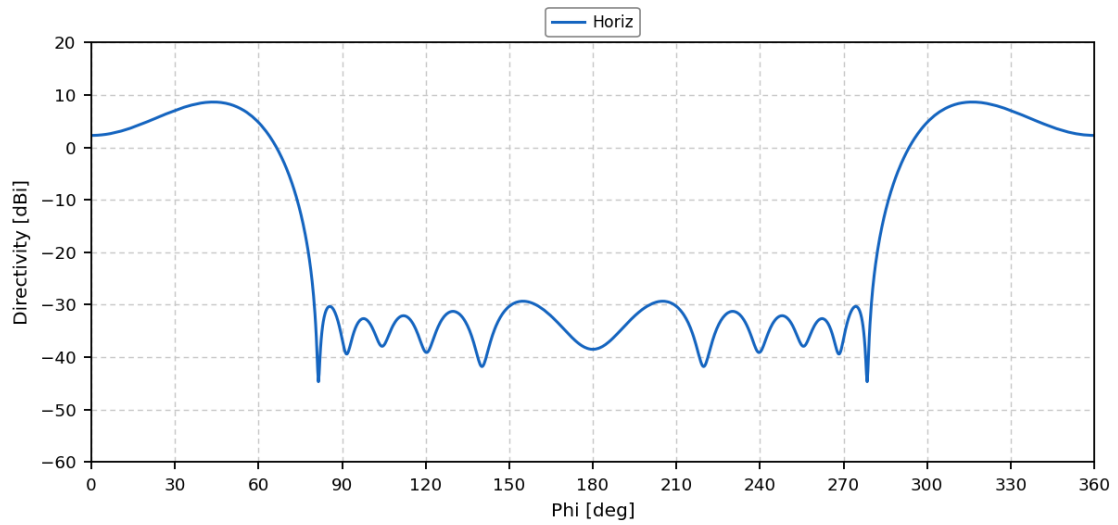
```
Tensiones para phi=45 grados:
1.0e+02 *

0.0568 + 0.0534i
0.1378 - 0.2926i
-0.7221 - 0.1027i
0.2019 + 1.1306i
1.1943 - 0.6738i
-1.0076 - 0.8146i
-0.1897 + 0.9592i
0.5166 - 0.2978i
```

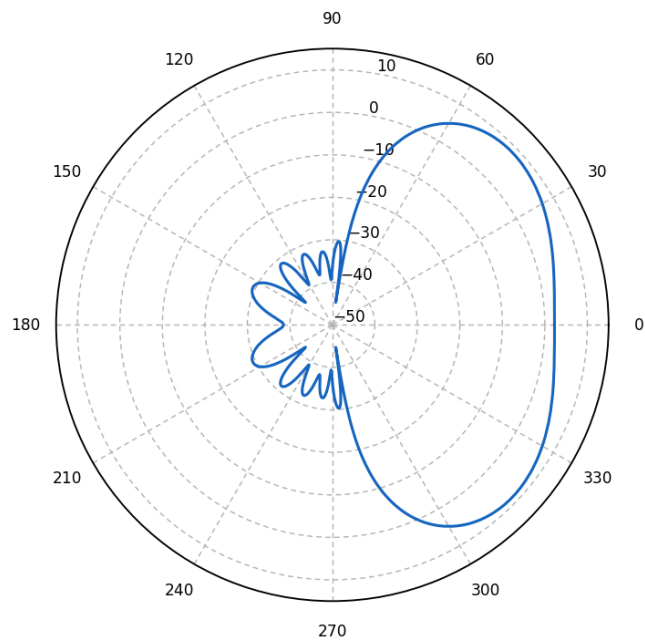
The following source settings are entered in the Prob4B_45 model:

n	Amp _n = V _n [V]	Afas _n = ∠V _n [°]
1	7,7974	43,2126
2	32,3424	-64,7734
3	72,9362	-171,9037
4	114,8455	79,8732
5	137,1259	-29,4289
6	129,5682	-141,0477
7	97,7772	101,1873
8	59,6296	-29,9584

Far field



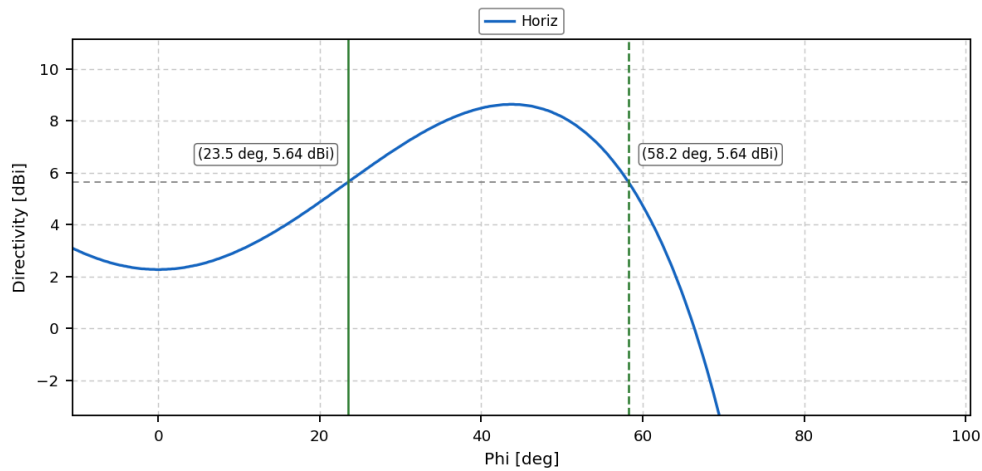
Total Directivity [dBi] (Frequency = 23 MHz; Theta = 90 deg) - Prob4B_45



Total Directivity (Frequency = 23 MHz; Theta = 90 deg) - Prob4B_45

Directividad = 8,62 dBi

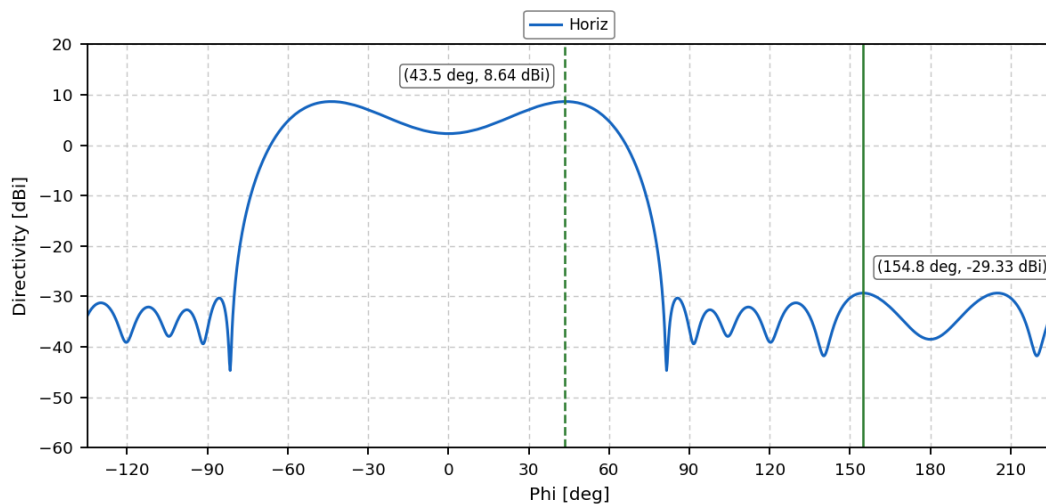
Far field



Total Directivity [dBi] (Frequency = 23 MHz; Theta = 90 deg) - Prob4B_45

Directivity = 8.62 dBi

Far field



Total Directivity [dBi] (Frequency = 23 MHz; Theta = 90 deg) - Prob4B_45

The maximum occurs at ϕ approximately 43.5° , essentially the intended direction. The apparent deviation is below 1.5° and results from the asymmetrical, very flat top of the lobe. Sidelobes separated by nulls remain 37.97 dB below the maximum. However, a new effect appears: the mirror beam at $\phi = -45^\circ$ merges with the main beam through a saddle at $\phi = 0^\circ$ only 6.37 dB below the peak. With $d = 0.4\lambda$ and steering to 45° , the visible region in ψ contains no null between the two beams, so the main lobe degenerates into a dual-beam pattern and the effective sidelobe ratio is limited by the saddle.

2.5.3. Comparison Table

	Cases	Directivity	Peak φ	Beamwidth	SLL
$\varphi = 0^\circ$		11,03 dBi	0°	$69,2^\circ$	32,20 dB
$\varphi = 45^\circ$		8,62 dBi	$\approx 44^\circ$	$34,8^\circ$	37,97 dB ⁽¹⁾
$\varphi = 90^\circ$		10,43 dBi	90°	$22,1^\circ$	37,22 dB

(1) Relative to sidelobes separated by nulls; the saddle connecting the $\pm 45^\circ$ beams is only 6.37 dB below the maximum.

Comments on the variations:

The most significant effect of scanning away from broadside is beam broadening. Angular resolution worsens as the beam moves away from $\varphi = 90^\circ$: HPBW increases from 22.1° to 34.8° at 45° and to 69.2° at endfire, because the projected array aperture decreases with $\sin(\varphi_0)$.

At $\varphi = 45^\circ$, directivity falls to 8.62 dBi. In addition to beam broadening, some power is radiated into the mirror beam at -45° and into the saddle joining the two beams. At $\varphi = 0^\circ$, directivity remains high and even rises to 11.03 dBi despite the broad beam, because the endfire beam is rotationally symmetric about the x-axis (a single conical pencil beam in 3D) rather than the bidirectional fan beam of the broadside case.

Sidelobe suppression is maintained with margin during scanning (37.2 to 38.0 dB), except for the endfire back lobe (32.2 dB) and the 45° saddle. The selected spacing $d = 0.40\lambda$ prevents grating lobes throughout the visible region.

3. YAGI

The element lengths and spacings are varied to meet the following requirements:

1. The driven element must be a folded dipole. The conductor spacing is $(0.2 C_3 + 1.5)d = 1.5d$, where d is the wire diameter:

$$s_a = 1,5 \cdot 0,06301 = 0,09452 \text{ m}$$

2. The antenna directivity must be > 9 dB.
3. The front-to-back ratio must be > 15 dB.
4. The input impedance must be real.

The following normalized dimensions are varied:

- Driven-element length (L_a).
- Director-element lengths (L_{d1} and L_{d2}).
- Reflector length (L_r).
- Spacing to the director elements (S_{d1} and S_{d2}).
- Spacing to the reflector (S_r).

OPTFEKO, using the Nelder-Mead simplex method, is applied in stages. One element is added in each model (Prob5A through Prob5E) to approach the specifications. The elements are parallel to the x-axis and stacked along z, so the beam points toward +z ($\theta = 0^\circ$).

3.1. Driven Element

First, L_a is optimized to obtain a real input impedance, using $\text{Im}(Z_{in}) = 0$ as the OPTFEKO objective in Prob5A. The optimized value is:
 $L_a = 0.4517187500\lambda = 5.8920 \text{ m}$.

```
Value of the aim function (analysis no. 26) is 6.742850890e-08

SIMPLEX NELDER-MEAD finished
Standard deviation small enough: 8.060475319e-07

Sensitivity of optimum value with respect to each optimisation parameter:
Parameter          Sensitivity
a_la                2.419798272e-02

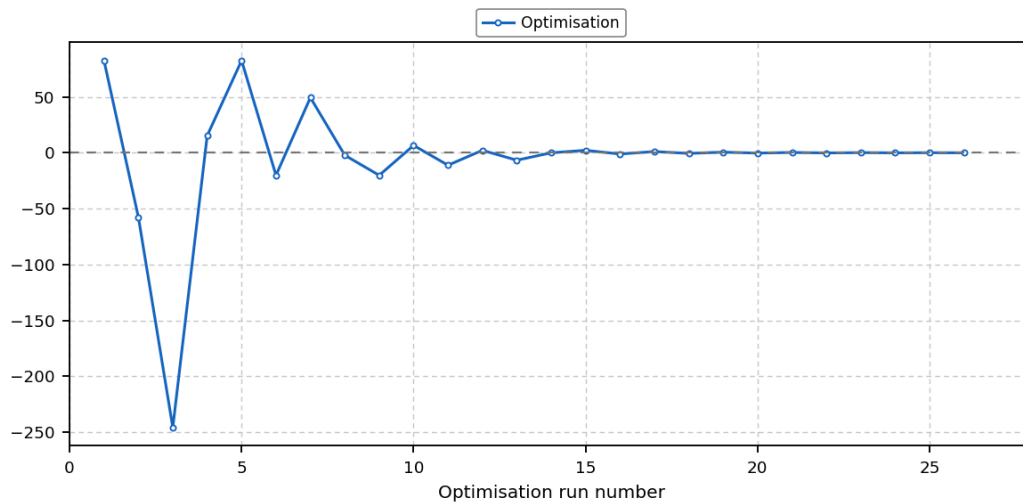
Optimum found for these variables:
a_la                = 4.517187500e-01
Optimum aim function value: 6.742850890e-08
```

After optimization, the resulting value is:

$$L_a = 4,517187500e-01 \lambda = 5,8920 \text{ m}$$

The convergence trace for the impedance objective, specifically the imaginary part of Z_{in} at each analysis step, is shown in the original-page panel.

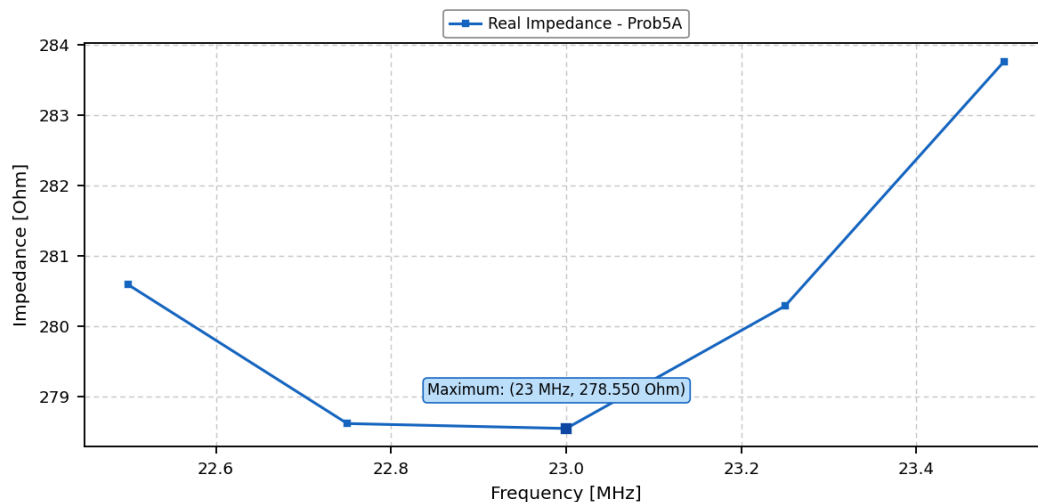
Optimisation



Trace = impedancia.goals.impedancegoal1

The optimized input impedance is $Z_{in} = 278.550 + j0.009 \Omega$, which is effectively real. The value is approximately four times 70Ω , as expected for a resonant folded dipole because of its 4:1 impedance transformation relative to a simple dipole.

Miscellaneous Data



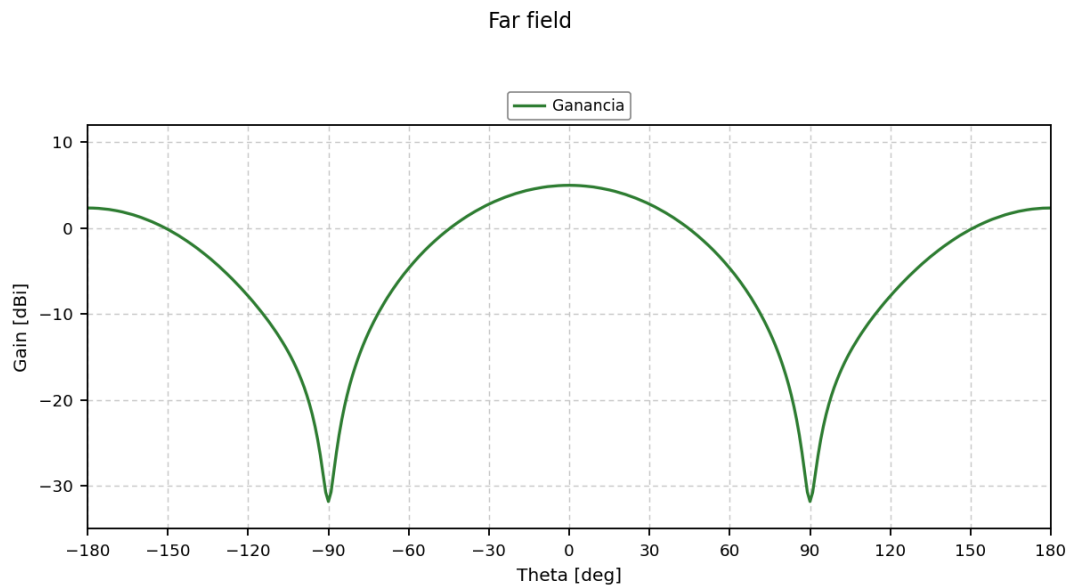
3.2. Director Element 1

With the driven element established, the first director length L_{d1} and spacing S_{d1} are optimized. The goal is to maintain a nearly real input impedance while assigning greater weight to a gain target of approximately 5 dB. Optimization of Prob5B gives:

$$L_{d1} = 4,115431570e-01 \lambda$$

$$S_{d1} = 2,372150904e-01 \lambda$$

For these values, the E-plane pattern ($\phi = 0^\circ$) already shows the beam tilting toward the director (+z):



$$\text{Gain} = 5,000 \text{ dBi}$$

$$Z = 202,080 - j0,034 \Omega$$

The director is slightly shorter than the driven element and is coupled at 0.237λ . It loads the folded dipole and reduces the input resistance from approximately 278Ω for the isolated driven element to about 202Ω , while keeping reactance nearly zero.

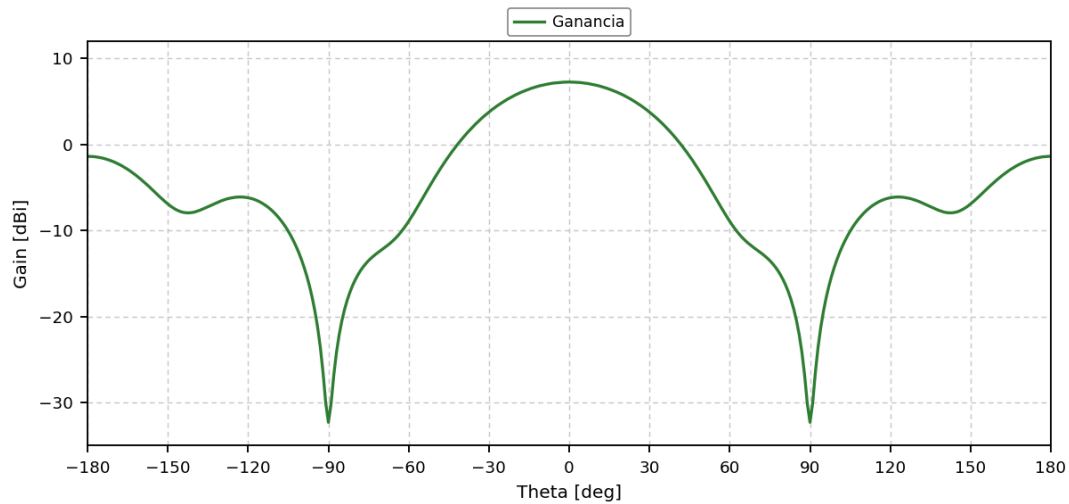
3.3. Director Element 2

The second director is introduced to increase gain while keeping the input impedance as close to real as possible. The second-director length $Ld2$ and spacing $Sd2$ are varied in Prob5C.

$$L_{d2} = 4,353182978e-01 \lambda$$

$$S_{d2} = 3,586019252e-01 \lambda$$

Far field



Total Gain [dBi] (Frequency = 23 MHz; Phi = 0 deg) - Prob5C

The obtained gain is:

$$\text{Gain} = 7,280 \text{ dBi}$$

$$L_{d2} = 0.4353182978\lambda$$

$$\mathbf{Z} = 168,440 - j47,104 \Omega$$

Gain increases by 2.28 dB relative to the previous stage, at the cost of a significant capacitive reactance. This reactance will be removed in the final step by readjusting L_a .

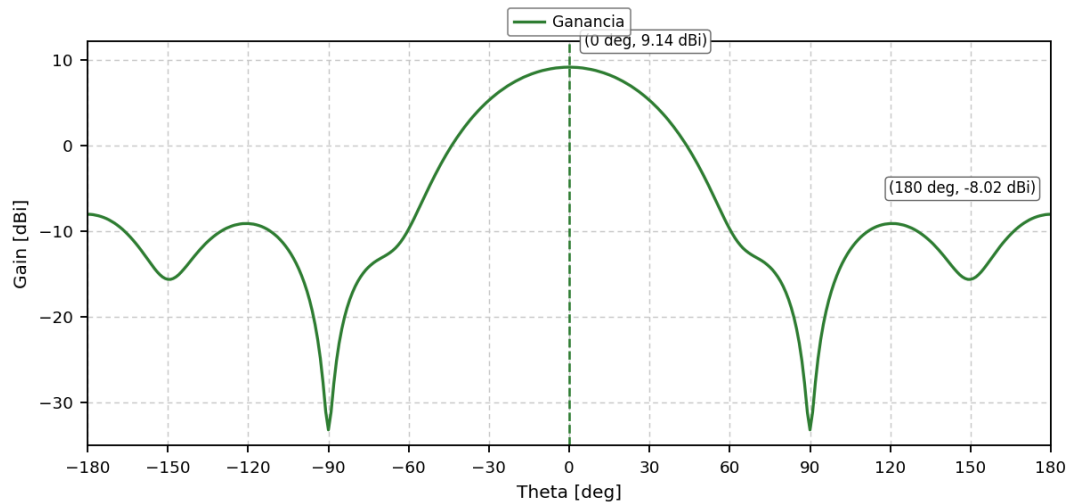
3.4. Reflector Element

The reflector is then added in Prob5D. Its length L_r and spacing S_r are optimized to approach the final requirements: gain near 9 dB, front-to-back ratio of at least 15 dB, and a nearly real input impedance.

$$L_r = 5,096777344e-01 \lambda$$

$$S_r = 2,150317383e-01 \lambda$$

Far field



Total Gain [dBi] (Frequency = 23 MHz; Phi = 0 deg) - Prob5D

The front-to-back ratio is:

$$\text{Relación delante-atrás} = 9,14 - (-8,02) = 17,16 \text{ dB}$$

With a gain of:

$$G = 9,140 \text{ dBi}$$

$$\text{Front-to-back ratio} = 9.14 - (-8.02) = 17.16 \text{ dB}$$

$$Z = 147,060 - j14,074 \Omega$$

3.5. COMPLETE ANTENNA: PARAMETER CALCULATION

A final joint optimization of all seven variables is performed in Prob5E. The simultaneous constraints are $D > 9 \text{ dB}$, $F/B > 15 \text{ dB}$, and $\text{Im}(Z_{in}) = 0$. The operating band is centered at $f_0 = 23 \text{ MHz}$ by slightly rescaling the dimensions and readjusting L_a , which primarily controls reactance.

$$L_a = 4,423514938e-01 \lambda$$

$$L_r = 4,776942401e-01 \lambda$$

$$S_r = 1,887796822e-01 \lambda$$

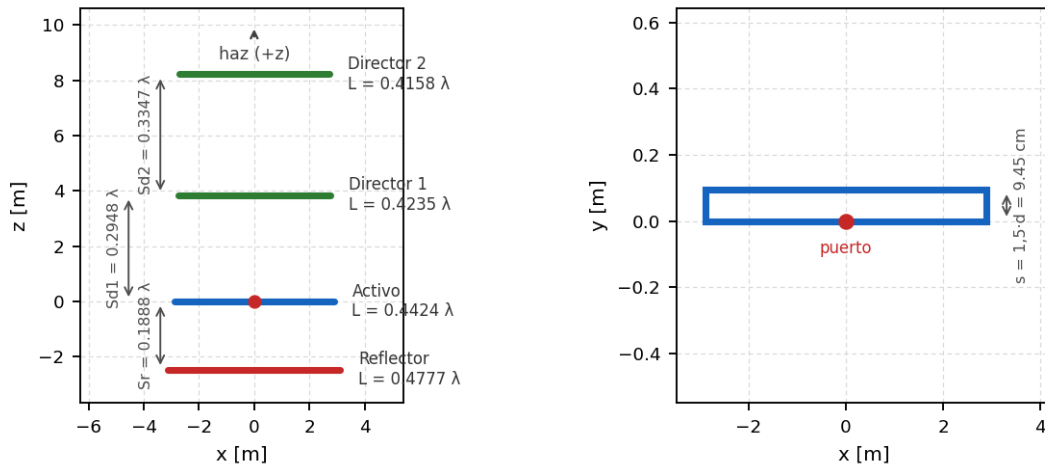
$$L_{d1} = 4,235114853e-01 \lambda$$

$$S_{d1} = 2,948081382e-01 \lambda$$

$$L_{d2} = 4,158418275e-01 \lambda$$

$$S_{d2} = 3,346865393e-01 \lambda$$

The total boom length is $S_r + S_{d1} + S_{d2} = 0.8183\lambda = 10.67$ m. The final geometry is shown in the original-page panel.



Yagi de 4 elementos (alzado x-z) y detalle del dipolo doblado (planta x-y, no a escala en y) - Prob5E

3.5.1. Directivity (Gain)

$$D = G = 10,410 \text{ dBi} > 9 \text{ dB} \rightarrow \text{CUMPLIMOS}$$

The antenna is lossless, so directivity and gain are identical. The maximum is at $\theta = 0^\circ$, toward the directors.

3.5.2. Input Impedance

$$Z = 90,789 - j0,004 \Omega \rightarrow \text{real} \rightarrow \text{CUMPLIMOS}$$

The folded dipole keeps the input resistance at a convenient value of approximately 91 Ω . Parasitic-element loading reduces the driven-element resistance, while the folded-dipole 4:1 transformation brings it back to an easily matched level.

3.5.3. Front-to-Back Ratio

$$F/B = 10.41 - (-8.00) = 18.41 \text{ dB} > 15 \text{ dB: requirement met.}$$

3.5.4. Bandwidth

The bandwidth is defined as the interval over which all design requirements are met simultaneously:

1. Matching: $VSWR < 2$, equivalent to $|S_{11}| < -9.5$ dB, referenced to $Z_0 = 90.79 \Omega$, the input resistance at f_0 .
2. Gain: directivity above 9 dB.
3. Pattern purity: front-to-back ratio above 15 dB.

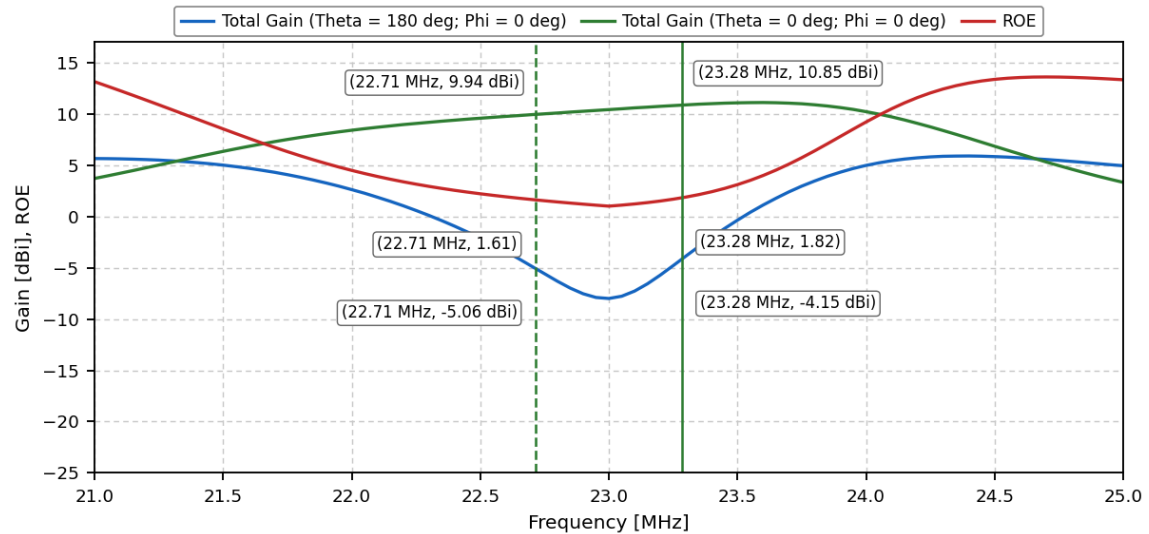
1. Matching: $VSWR < 2$ (equivalent to $|S_{11}| < -9.5$ dB), referred to $Z_0 = 90.79 \Omega$, the input resistance at f_0 . This standard criterion ensures efficient power transfer and protects the transmitter.

2. Gain: directivity greater than 9 dB, maintaining the original design requirement.

3. Main-lobe quality: front-to-back ratio greater than 15 dB to ensure rejection of rear interference.

A frequency sweep from 21 to 25 MHz is performed using Prob5EFrec. The curves show forward gain ($\theta = 0^\circ$), back radiation ($\theta = 180^\circ$), and VSWR.

Far field



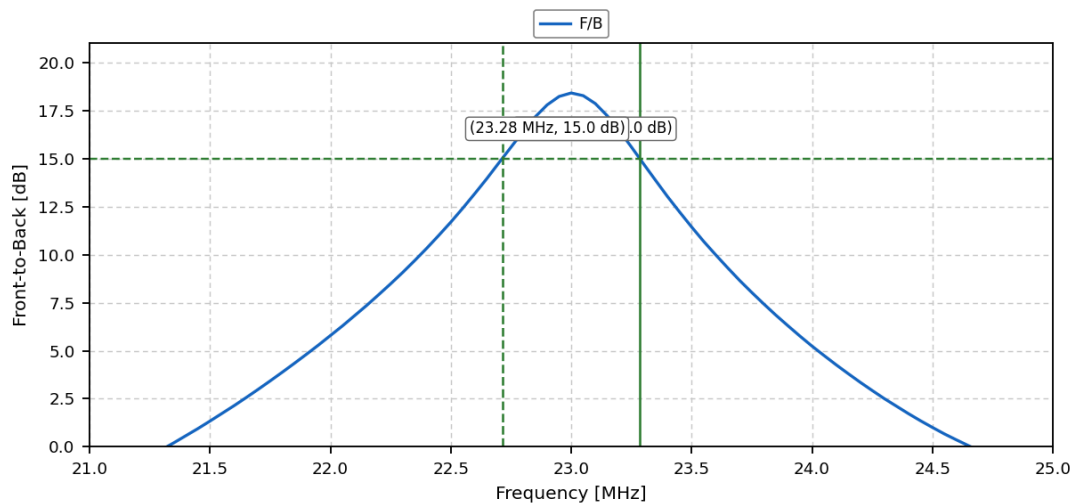
Total Gain / ROE (Zref = 90.8 Ohm) - Prob5EFrec

Individual criteria:

	Criterio fmín [MHz]	fmax [MHz]	Margen [MHz]
D > 9 dB	22,223	24,210	1,988
F/B > 15 dB	22,714	23,284	0,571
ROE < 2	22,569	23,324	0,755

The front-to-back ratio is the most restrictive parameter and sets both band edges:

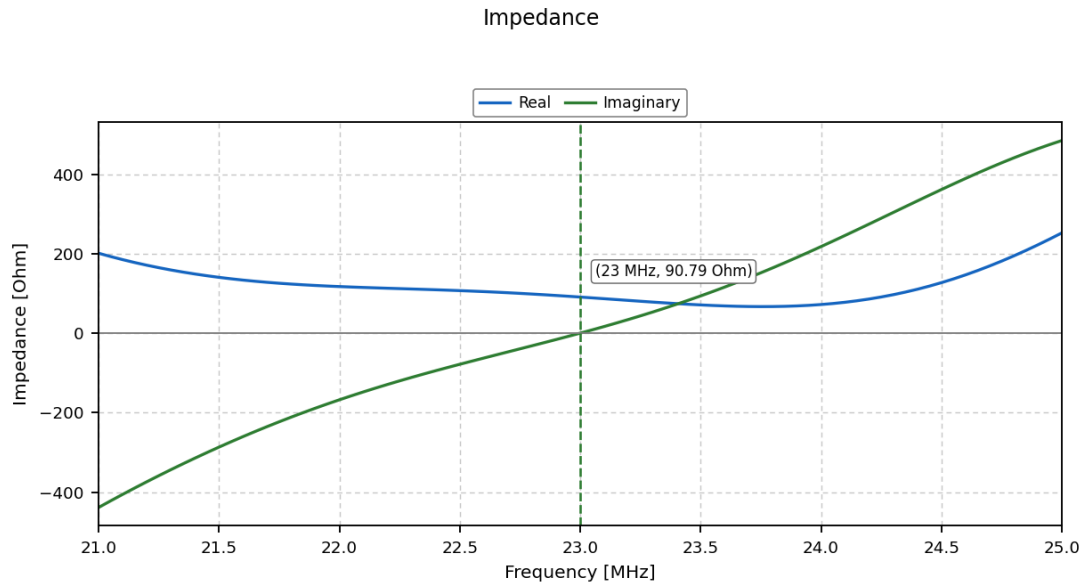
Far field



Relacion delante-atras - Prob5EFrec

fmin = 22.714 MHz to fmax = 23.284 MHz -> BW = 0.571 MHz (2.48% of f0)

The input-impedance curve confirms the resonant behavior of the driven element: $\text{Im}(Z)$ is exactly zero at f_0 and changes smoothly across the band.



Impedance (Frequency sweep) - Prob5EFrec

Within the calculated bandwidth, all requested parameters are evaluated:

f [MHz]	D [dBi]	F/B [dB]	Z_{in} [Ω]	ROE	HPBW _E	HPBW _H	NLPS [dB] _E	NLPS [dB] _H
22,71	9,93	14,93	101,3 – j45,7	1,62	54,6°	71,2°	14,9	12,5
22,85	10,17	17,16	96,6 – j24,0	1,30	53,9°	69,8°	17,2	13,8
23,00	10,41	18,41	90,8 – j0,0	1,00	53,2°	68,2°	18,4	14,9
23,15	10,65	17,22	84,5 + j25,4	1,35	52,2°	66,4°	17,2	14,3
23,28	10,85	15,08	79,0 + j49,1	1,80	51,3°	64,6°	15,1	13,7

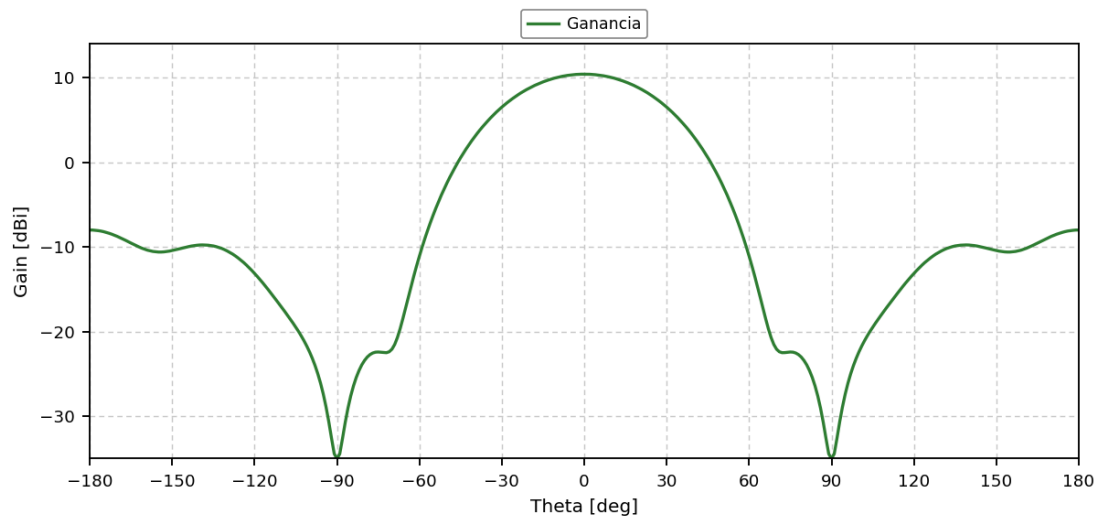
At the band edges, the limiting F/B value is approximately 15 dB. Directivity retains more than 0.9 dB of margin, and VSWR remains below 1.8. In both principal planes, the largest sidelobe is the back lobe, so its variation follows the front-to-back ratio.

3.5.5. RADIATION PATTERNS IN THE PRINCIPAL PLANES

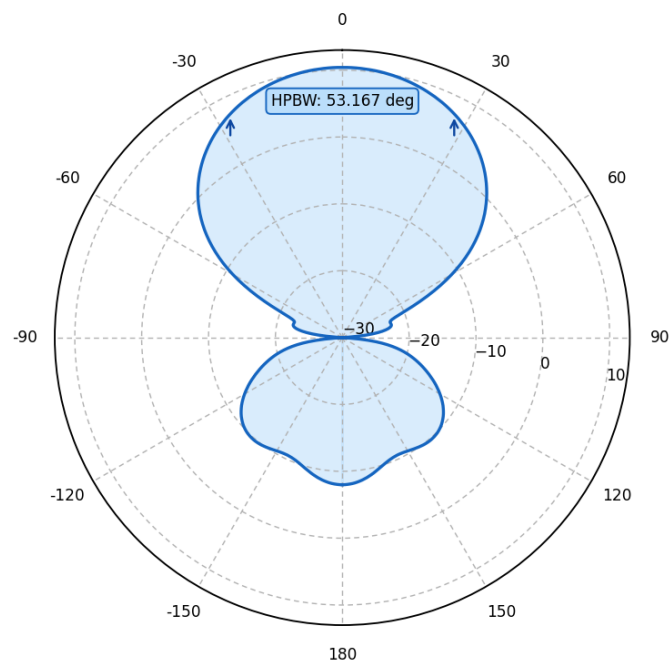
The antenna principal planes are:

- E-plane: $\phi = 0^\circ$, the x-z plane containing the element arms and the boom;
- H-plane: $\phi = 90^\circ$, the y-z plane.

Far field



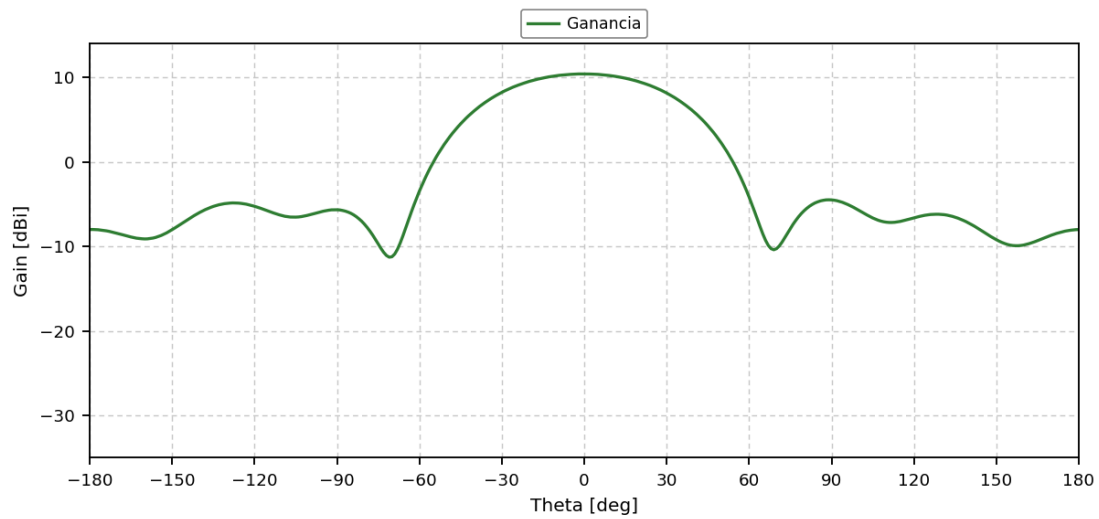
Total Gain [dBi] (Frequency = 23 MHz; Phi = 0 deg) - Prob5E



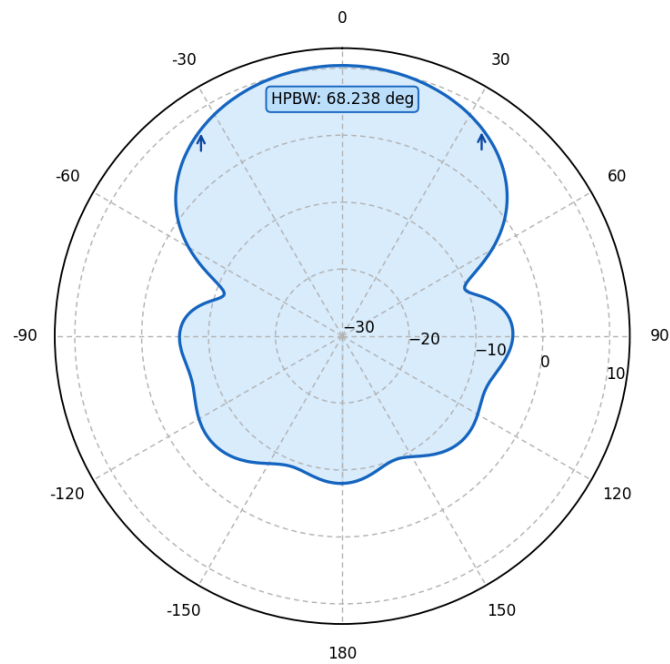
Total Gain [dBi] (Frequency = 23 MHz; Phi = 0 deg) - Prob5E

The E-plane half-power beamwidth is 53.17° . The nulls at $\theta = \pm 90^\circ$ are the intrinsic dipole nulls in the direction of the arms.

Far field



Total Gain [dBi] (Frequency = 23 MHz; Phi = 90 deg) - Prob5E

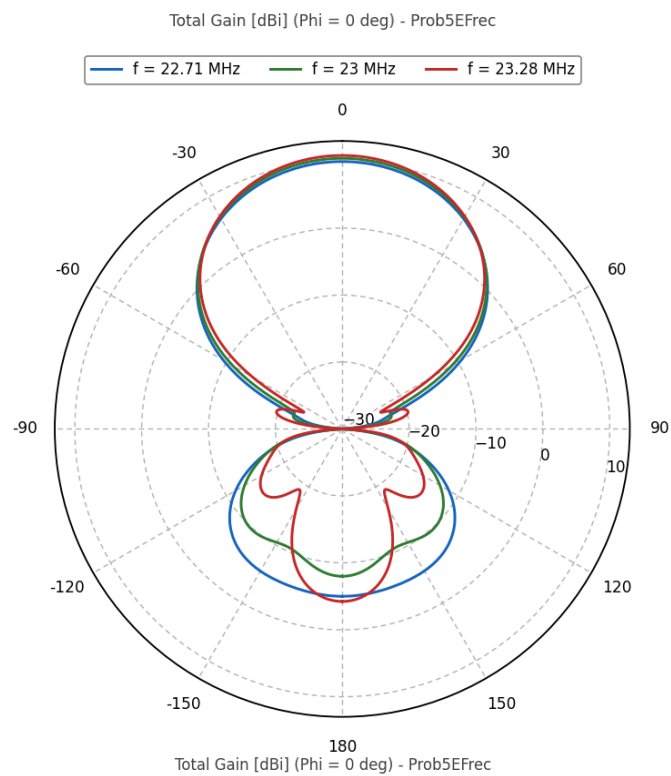
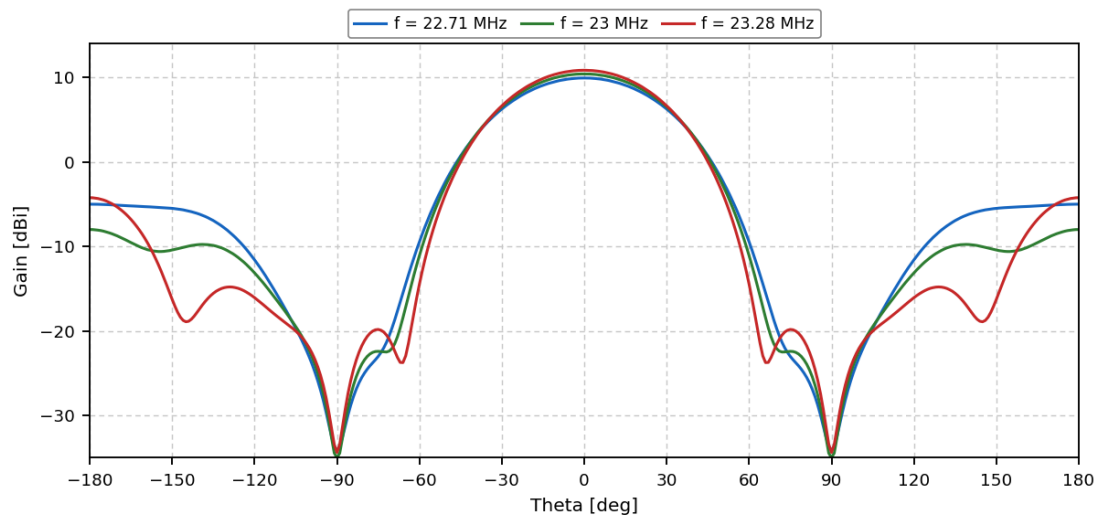


Total Gain [dBi] (Frequency = 23 MHz; Phi = 90 deg) - Prob5E

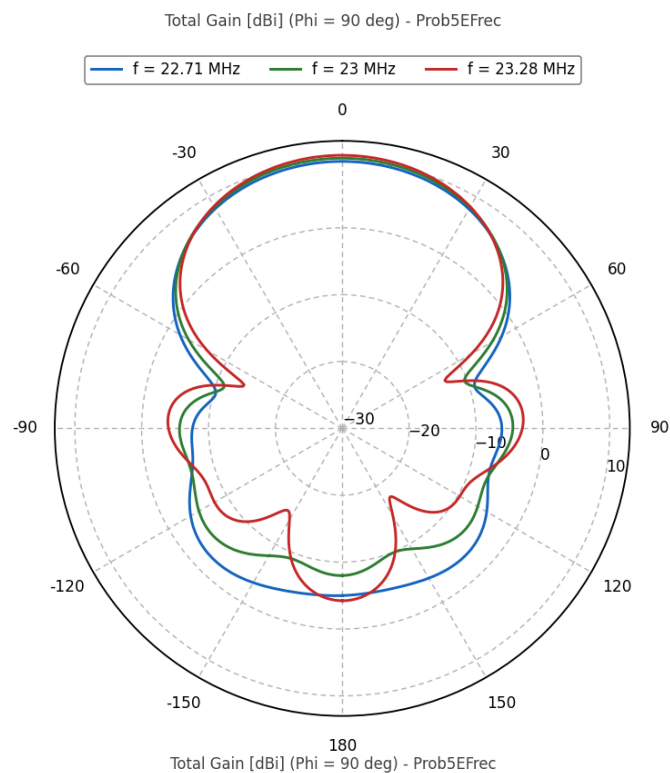
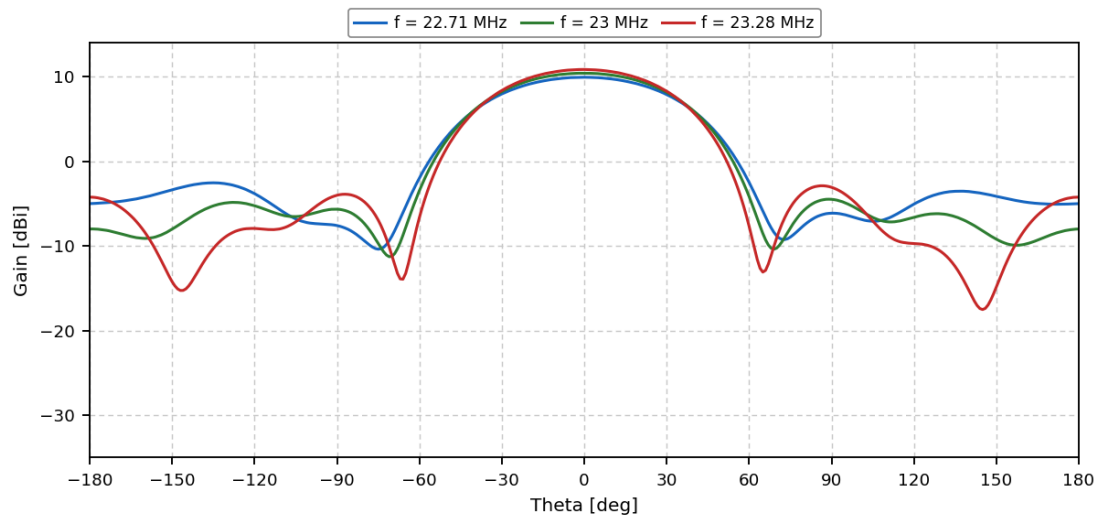
In the H-plane ($\phi = 90^\circ$), the beamwidth is 68.24° . The H-plane beam is wider because the elements are aligned along z and there is no stacking in the y direction; directivity in this plane is produced only by the director-reflector action.

Both principal-plane cuts are repeated at the center frequency and at the calculated band edges: 22.71, 23.00, and 23.28 MHz. The overlaid patterns are shown in the original-page panel.

Far field



Far field



The main lobe changes very little across the operating band. Beamwidth varies by less than 4° in either principal plane, and the maximum remains at $\theta = 0^\circ$. The most visible change is the growth of the back lobe near the band edges, consistent with the front-to-back ratio being the parameter that limits the antenna bandwidth.